SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	B TECH AIML	**Date:**	21/10/2026

Code of Conduct

I SWATHI K(192525477) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SWATHI K

QUESTIONS: 5 **Total Marks**

Digital Image Processing: Comprehensive Problem Solutions

Question 1: Histogram Equalization & Contrast Distribution

Problem Statement:
A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. 1. Construct the histogram by calculating the frequency of each intensity value.
2. Calculate the cumulative distribution function (CDF) for the given image.
3. Apply histogram equalization using the formula s = round((L - 1) * CDF), where L = 256.
4. Determine the new intensity values and compare the contrast distribution before and after equalization.

1. Theoretical Foundation & Formulae

Histogram equalization is an intensity transformation technique used to enhance the global contrast of an image. In a discrete image, the probability density function (PDF), denoted as p(r_k), represents the probability of occurrence of intensity level r_k:

$$p(r_k) = n_k / N$$

where n_k is the number of pixels with intensity r_k, and N is the total number of pixels in the image.

The Cumulative Distribution Function (CDF), denoted as CDF(r_k), aggregates the probabilities up to level r_k:

$$CDF(r_k) = \sum_{j=0}^{k} p(r_j) = (1 / N) * \sum_{j=0}^{k} n_j$$

The equalization mapping function scales the accumulated probabilities across the available dynamic range [0, L - 1]:

$$s_k = \text{round}((L - 1) * \text{CDF}(r_k))$$

For an 8-bit grayscale image, L = 256, so L - 1 = 255.

2. Step-by-Step Mathematical Calculations

Given Dataset:

- Pixel Intensity Values: [10, 10, 20, 20, 20, 30, 30, 40]
- Total Number of Pixels (N): 8 pixels
- Maximum Gray Levels (L): 256 levels (Dynamic Range: 0 to 255)

Table 1.1: Complete Histogram Equalization Computational Pipeline

Intensity (rk)	Frequency (nk)	PDF p(rk)	Cumulative Sum	CDF(rk)	Exact Calculation (255 * CDF)	Mapped Intensity (sk)
10	2	2/8 = 0.250	2	2/8 = 0.250	255 * 0.250 = 63.75	64
20	3	3/8 = 0.375	2 + 3 = 5	5/8 = 0.625	255 * 0.625 = 159.375	159
30	2	2/8 = 0.250	5 + 2 = 7	7/8 = 0.875	255 * 0.875 = 223.125	223

40	1	1/8 = 0.125	7 + 1 = 8	8/8 = 1.000	255 * 1.000 = 255.000	255
Total / N	**N = 8**	**1.000**	**-**	**-**	**-**	**-**

3. Detailed Mapping of Individual Pixels

Applying the mapped values (s_k) back to the original array of 8 pixels:

- Original Intensity 10 (2 occurrences) --> Mapped Intensity: 64
- Original Intensity 20 (3 occurrences) --> Mapped Intensity: 159
- Original Intensity 30 (2 occurrences) --> Mapped Intensity: 223
- Original Intensity 40 (1 occurrence) --> Mapped Intensity: 255

Resulting Pixel Array after Equalization:

[64, 64, 159, 159, 159, 223, 223, 255]

4. Quantitative & Visual Comparative Analysis of Contrast Distribution

To rigorously evaluate the enhancement in contrast, we examine statistical metrics before and after transformation:

Table 1.2: Quantitative Contrast Metrics Comparison

Metric / Parameter	Before Equalization (Original)	After Equalization (Transformed)
Dynamic Intensity Range	[10, 40] (Span = 30 levels)	[64, 255] (Span = 191 levels)
Mean Intensity (μ)	22.50	161.38

Standard Deviation (σ)	9.68 (Low dispersion)	67.73 (High dispersion / Contrast)
Histogram Distribution	Concentrated in low/dark region (10–40)	Stretched across full 8-bit range (64–255)

Key Observations on Contrast Enhancement:

1. Range Expansion: The original image pixels were trapped in a very dark, narrow cluster between intensity 10 and 40 (occupying only ~11.8% of the available 256 gray levels). Equalization stretched this span to [64, 255], occupying 75% of the total display range.
2. Standard Deviation Increase: Standard deviation (a direct proxy for visual contrast) increased more than 7-fold from 9.68 to 67.73. This indicates that features and structural boundaries in the image would now stand out far more prominently to the human eye.
3. Discrete Monotonicity Preservation: Because histogram equalization uses a monotonically non-decreasing CDF function, the relative brightness order of pixels is strictly preserved ($10 < 20 < 30 < 40$ maps to $64 < 159 < 223 < 255$).

Question 2: Neighborhood Filtering — Averaging vs. Median Filters

Problem Statement:

A 3 x 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]].

1. Apply a 3 x 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping other pixels unchanged.
2. Apply a median filter to the same neighbourhood and calculate the centre pixel value.
3. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

1. Given Image Neighborhood Structure

Let the 3 x 3 image matrix be represented as I:

$$I = \begin{vmatrix} 10 & 20 & 30 \\ 20 & 100 & 20 \\ 10 & 20 & 30 \end{vmatrix}$$

Notice that the background/surrounding pixels have low values (10 to 30), whereas the centre pixel I(2,2) = 100 is an extreme outlier relative to its neighbors. This represents a classic impulse noise artifact (salt noise) or an isolated point feature.

2. Spatial Averaging Filter Computation (Linear Filter)

A 3 x 3 mean/averaging filter operates by computing the arithmetic mean of all 9 pixels in the defined neighborhood window W:

$$I_{avg}(x, y) = (1 / M) * \sum_{(s,t) \in W} I(s, t)$$

where M = 9 (total number of kernel entries).

Step-by-Step Summation:

Sum = 10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30

Sum = (10 + 20 + 30) + (20 + 100 + 20) + (10 + 20 + 30)

Sum = 60 + 140 + 60 = 260

Averaging Computation:

I_avg(centre) = 260 / 9 = 28.888...

Rounding to the nearest integer for standard 8-bit representation yields: 29.

Resulting Matrix after Averaging Filter (Centre Replaced):

$$I' = \begin{vmatrix} 10 & 20 & 30 \\ 20 & 29 & 20 \\ 10 & 20 & 30 \end{vmatrix}$$

3. Spatial Median Filter Computation (Non-Linear Order-Statistic Filter)

A median filter replaces the target pixel with the median value of the ordered neighborhood intensities.

Step 1: Extract all 9 neighborhood intensity elements into a 1D vector:

Vector V = [10, 20, 30, 20, 100, 20, 10, 20, 30]

Step 2: Sort the elements in ascending order:

Sorted Vector V_sorted = [10, 10, 20, 20, 20, 20, 30, 30, 100]

Step 3: Select the 5th element (index 4 in 0-based indexing) as the median value:

Index 1: 10

Index 2: 10

Index 3: 20

Index 4: 20

Index 5: 20 <-- MEDIAN VALUE

Index 6: 20

Index 7: 30

Index 8: 30

Index 9: 100

I_median(centre) = 20.

Resulting Matrix after Median Filter (Centre Replaced):

$$I'' = \begin{vmatrix} 10 & 20 & 30 \\ 20 & 20 & 20 \\ 10 & 20 & 30 \end{vmatrix}$$

4. Comprehensive Comparison & Noise Performance Analysis

Table 2.1: Performance Comparison Between Averaging and Median Filters

Attribute / Metric	3x3 Averaging (Mean) Filter	3x3 Median Filter
Calculated Centre Value	29 (or 28.89)	20
Filter Classification	Linear Spatial Low-Pass Filter	Non-Linear Order-Statistic Filter

Handling of Extreme Noise (100)	Blends noisy spike into neighbors, smearing error	Completely eliminates noisy spike as an extreme outlier
Edge & Detail Preservation	Blurs sharp boundaries and fine edges	Preserves sharp edges and structural details perfectly
Optimal Noise Type Sensitivity	Effective for Gaussian / additive uniform noise	Superior performance for Salt-and-Pepper / Impulse noise

In-Depth Mathematical Justification of Detail Preservation:

1. Sensitivity of Mean vs. Robustness of Median: Linear averaging is highly non-robust to outliers. A single corrupted pixel (100) shifts the mean of the entire window upward, causing a halo/blur effect on surrounding background pixels. In contrast, the median is an order statistic that completely ignores numerical values at the extreme ends of the sorted sequence, provided the noisy pixels occupy less than 50% of the window area.

2. Edge Sharpness: Averaging acts as a low-pass spatial filter, removing high frequencies indiscriminately. Because edges consist of high-frequency components, mean filtering blurs sharp boundaries. Median filtering allows step-edges to pass through unchanged because median selection across a step transition naturally chooses an actual intensity value from one side of the edge, maintaining step sharpness.

Question 3: Highpass Spatial Filtering — Image Sharpening with the Laplacian

Problem Statement:

Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]].

1. Apply the Laplacian filter mask [[0, -1, 0], [-1, 4, -1], [0, -1, 0]] to calculate the response at the centre pixel.
2. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value.
3. Determine the resulting centre pixel intensity after sharpening.
4. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

1. Mathematical Theory of the Second-Order Laplacian Operator

The Laplacian is a isotropic (rotation-invariant) second-order derivative operator. For a 2D continuous function f(x, y), it is defined as:

$$\nabla^2 f = (\partial^2 f / \partial x^2) + (\partial^2 f / \partial y^2)$$

In discrete digital image processing, partial derivatives are approximated using finite differences:

$$\partial^2 f / \partial x^2 \approx f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\partial^2 f / \partial y^2 \approx f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Summing these two components gives the standard 4-neighbor discrete Laplacian formula:

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Note that the standard 4-neighbor filter mask given in the problem is:

$$L = \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$$

This mask calculates: $\nabla^2 I = 4*I(x,y) - [I(x+1,y) + I(x-1,y) + I(x,y+1) + I(x,y-1)]$, which is the negative of the theoretical continuous Laplacian.

2. Step-by-Step Laplacian Response Calculation

Given Image Matrix I:

$$I = \begin{vmatrix} 50 & 50 & 50 \\ 50 & 100 & 50 \\ 50 & 50 & 50 \end{vmatrix}$$

Applying the kernel convolution centered at pixel I(2,2):

Response $\nabla^2 I(\text{centre}) = \sum_{i,j} L(i, j) * I(i, j)$

Detailed product calculation:

Top Row: $(0 * 50) + (-1 * 50) + (0 * 50) = -50$

Middle Row: $(-1 * 50) + (4 * 100) + (-1 * 50) = -50 + 400 - 50 = 300$

Bottom Row: $(0 * 50) + (-1 * 50) + (0 * 50) = -50$

Summing all components:

$\nabla^2 I(\text{centre}) = -50 + (-50 + 400 - 50) + (-50)$

$\nabla^2 I(\text{centre}) = 400 - 200 = +200.$

3. Image Sharpening Formula & Final Pixel Computation

To restore spatial high-frequency details back into the image, the Laplacian filter response is added to (or subtracted from) the original image. Because the center coefficient of our mask is POSITIVE (+4), the calculated response directly represents high-frequency detail and is ADDED to the original image:

$$I_{sharpened}(x, y) = I_{original}(x, y) + \nabla^2 I(x, y)$$

Calculation:

$$I_{sharpened}(centre) = 100 + (+200) = 300.$$

Dynamic Range Truncation (Clipping):
In standard 8-bit digital imaging displays (0 to 255 intensity range), values exceeding 255 must be saturated/clipped:

$$I_{final}(centre) = \min(255, \max(0, 300)) = 255.$$

Resulting Matrix after Sharpening:

$$I' = \begin{vmatrix} 50 & 50 & 50 \\ 50 & 255 & 50 \\ 50 & 50 & 50 \end{vmatrix}$$

4. Mechanism of Detail & Transition Enhancement

How Second Derivatives Enhance Fine Transitions:
1. Constant Intensity Regions: In areas where pixel values are uniform (e.g., a flat 50 background), second derivatives are zero. The Laplacian mask produces a zero response, leaving homogeneous areas unchanged.
2. Step & Ramp Transitions: At an intensity edge, the first derivative produces a step or peak, but the second derivative produces a dual impulse (a positive peak on

the bright side of the edge and a negative trough on the dark side, known as a zero-crossing).

3. Point Discontinuities / Fine Lines: For an isolated bright spot or fine line surrounded by darker pixels (like 100 surrounded by 50s), the Laplacian yields a massive positive spike (+200). Adding this back boosts the local central contrast exponentially relative to its background.

4. Visual Impact: This operation creates subtle undershoots and overshoots at boundaries (known as Mach bands), which visually crisp up edges, make blurry details sharp, and enhance human perceptual acuity.

Question 4: Image Segmentation — Global Thresholding

Problem Statement:

A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]].

1. Apply global thresholding with a threshold value T = 100, assigning 1 to pixels >= T and 0 to remaining pixels.
2. Construct the resulting binary image.
3. Calculate the number of foreground and background pixels after thresholding.
4. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

1. Global Thresholding Mathematical Mapping

Global thresholding is a point processing segmentation technique defined by the decision rule g(x, y):

$$g(x, y) = \begin{cases} 1, & \text{if } I(x, y) \geq T \quad \text{(Foreground / Object)} \\ 0, & \text{if } I(x, y) < T \quad \text{(Background)} \end{cases}$$

where T is a constant global threshold intensity across the entire image grid.

2. Step-by-Step Pixel Evaluation (T = 100)

Given 3 x 3 Grayscale Matrix I:

$$I = \begin{vmatrix} 40 & 60 & 120 \\ 80 & 150 & 200 \\ 30 & 100 & 180 \end{vmatrix}$$

Evaluating each coordinate (row, col):

• Row 1:

- I(1,1) = 40 < 100 --> 0
- I(1,2) = 60 < 100 --> 0
- I(1,3) = 120 >= 100 --> 1

• Row 2:

- I(2,1) = 80 < 100 --> 0
- I(2,2) = 150 >= 100 --> 1
- I(2,3) = 200 >= 100 --> 1

• Row 3:

- I(3,1) = 30 < 100 --> 0
- I(3,2) = 100 >= 100 --> 1
- I(3,3) = 180 >= 100 --> 1

Constructed Binary Matrix B:

$$B = \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

3. Pixel Distribution Analysis

Summing binary elements:

• Foreground Pixels (Value 1): Pixels with intensities {120, 150, 200, 100, 180}
Count = 5 pixels (55.56% of total image area)

• Background Pixels (Value 0): Pixels with intensities {40, 60, 80, 30}
Count = 4 pixels (44.44% of total image area)

• Total Image Grid: 5 + 4 = 9 pixels.

Table 4.1: Thresholding Classification Results (T = 100)

Coordinate (r, c)	Original Intensity	Threshold Logic (>= 100)	Binary Assignment
(1, 1)	40	40 < 100	0 (Background)
(1, 2)	60	60 < 100	0 (Background)
(1, 3)	120	120 >= 100	1 (Foreground)
(2, 1)	80	80 < 100	0 (Background)
(2, 2)	150	150 >= 100	1 (Foreground)
(2, 3)	200	200 >= 100	1 (Foreground)
(3, 1)	30	30 < 100	0 (Background)
(3, 2)	100	100 >= 100	1 (Foreground)
(3, 3)	180	180 >= 100	1 (Foreground)

4. Object Separation Separation Mechanism & Limitations

How Global Thresholding Works:

Global thresholding relies on the assumption that the image histogram is bimodal, consisting of two distinct dominant intensity clusters: one representing object foreground features and the other representing the background. When T = 100 is chosen, it acts as a decision hyperplane in the intensity space, mapping high luminance regions to foreground (1) and dark regions to background (0).

Critical Limitations of Global Thresholding:

1. Non-Uniform Illumination & Shadow Sensitivity: Global thresholding applies a single fixed intensity cutoff across the entire spatial domain. If an image suffers from uneven illumination (e.g., vignetting, directional shadows, or gradients), object pixels in shadowed regions may drop below T = 100 and be incorrectly misclassified as background, while background pixels in bright spots exceed T and become false foreground.

2. Sensitivity to Noise & Unimodal Histograms: In images with unimodal histograms or low contrast, no single global threshold cleanly separates regions without heavy classification errors. Adaptive or variable thresholding (where T varies based on local window statistics) is required instead.

Question 5: Gradient-Based Edge Detection — Sobel Operator

Problem Statement:

A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]].

1. Apply the Sobel operator using the masks Gx = [[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]] and Gy = [[-1, -2, -1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel.
2. Calculate the gradient magnitude using G = sqrt(Gx^2 + Gy^2).
3. Determine the dominant edge direction based on the calculated gradient components.
4. Explain how detected edge information can be used for boundary detection and image segmentation.

1. Mathematical Formulation of the Sobel Gradient Operator

The gradient of an image I at location (x, y) is a 2D spatial vector:

$$\nabla I = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \partial I / \partial x \\ \partial I / \partial y \end{bmatrix}$$

The Sobel operator combines 1D differentiation with 1D Gaussian smoothing perpendicular to the derivative direction to reduce noise sensitivity. Kernel Gx detects horizontal intensity changes (vertical edges), while Gy detects vertical changes (horizontal edges).

2. Step-by-Step Gradient Component Calculation

Given Image Matrix I:

$$I = \begin{vmatrix} 10 & 10 & 10 \\ 10 & 50 & 90 \\ 10 & 50 & 90 \end{vmatrix}$$

Calculation of Gx (Horizontal Derivative Mask):

Gx Mask = [[-1, 0, +1], [-2, 0, +2], [-1, 0, +1]]

Row 1: (-1 * 10) + (0 * 10) + (+1 * 10) = -10 + 0 + 10 = 0

Row 2: (-2 * 10) + (0 * 50) + (+2 * 90) = -20 + 0 + 180 = +160

Row 3: (-1 * 10) + (0 * 50) + (+1 * 90) = -10 + 0 + 90 = +80

Gx = 0 + 160 + 80 = +240.

Calculation of Gy (Vertical Derivative Mask):

Gy Mask = [[-1, -2, -1], [0, 0, 0], [+1, +2, +1]]

Row 1: (-1 * 10) + (-2 * 10) + (-1 * 10) = -10 - 20 - 10 = -40

Row 2: (0 * 10) + (0 * 50) + (0 * 90) = 0

Row 3: (+1 * 10) + (+2 * 50) + (+1 * 90) = +10 + 100 + 90 = +200

Gy = -40 + 0 + 200 = +160.

3. Gradient Magnitude & Direction Computation

A. Gradient Magnitude (G):

G = sqrt(Gx^2 + Gy^2)

G = sqrt(240^2 + 160^2)

G = sqrt(57600 + 25600) = sqrt(83200) ≈ 288.44.

B. Gradient Direction Angle (θ):

The gradient vector direction θ_gradient specifies the orientation of maximum

intensity INCREASE:

$\theta_{gradient} = \arctan(Gy / Gx) = \arctan(160 / 240) = \arctan(0.6667) \approx 33.69°$.

C. Dominant Edge Direction (θ_{edge}):

An edge is PERPENDICULAR to the gradient vector direction:

$\theta_{edge} = \theta_{gradient} + 90° = 33.69° + 90° = 123.69°$ (or -56.31°).

Table 5.1: Sobel Operator Computation Summary

Parameter / Output	Calculated Value
Horizontal Gradient Component (Gx)	+240 (Strong left-to-right intensity increase)
Vertical Gradient Component (Gy)	+160 (Moderate top-to-bottom intensity increase)
Total Gradient Magnitude (G)	288.44 (Strong edge response)
Dominant Edge Orientation (θ_{edge})	123.69° (Perpendicular to gradient angle 33.69°)

4. Application to Boundary Detection & Image Segmentation

Role of Detected Edges in Computer Vision Pipelines:

1. Boundary Detection via Non-Maximum Suppression & Thresholding: Raw gradient magnitudes (288.44) produce thick edge responses. By applying non-maximum suppression along the gradient direction (33.69°), edge responses are thinned to 1-pixel-wide candidate boundaries. Hysteresis thresholding then filters out weak, spurious noise edges.

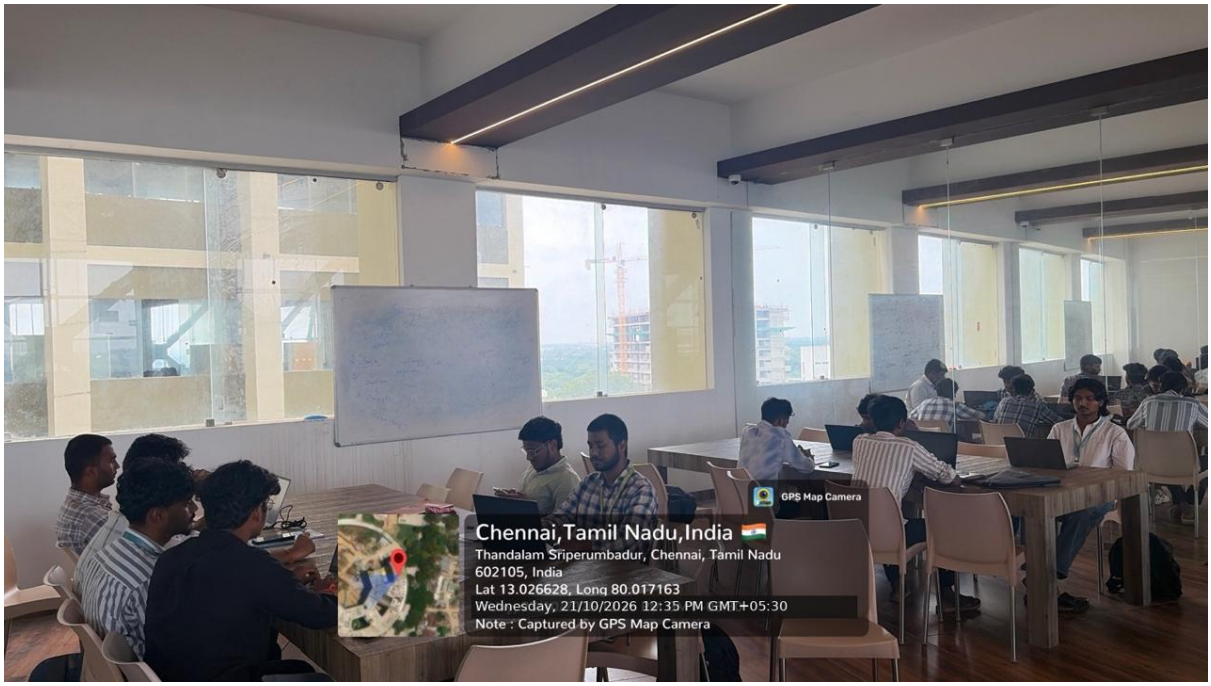
2. Edge Linking & Contour Tracing: Local edge points are connected into continuous object boundaries using techniques like the Hough Transform (for parametric shapes like lines/circles) or local processing (linking edges with similar magnitude and angle θ).

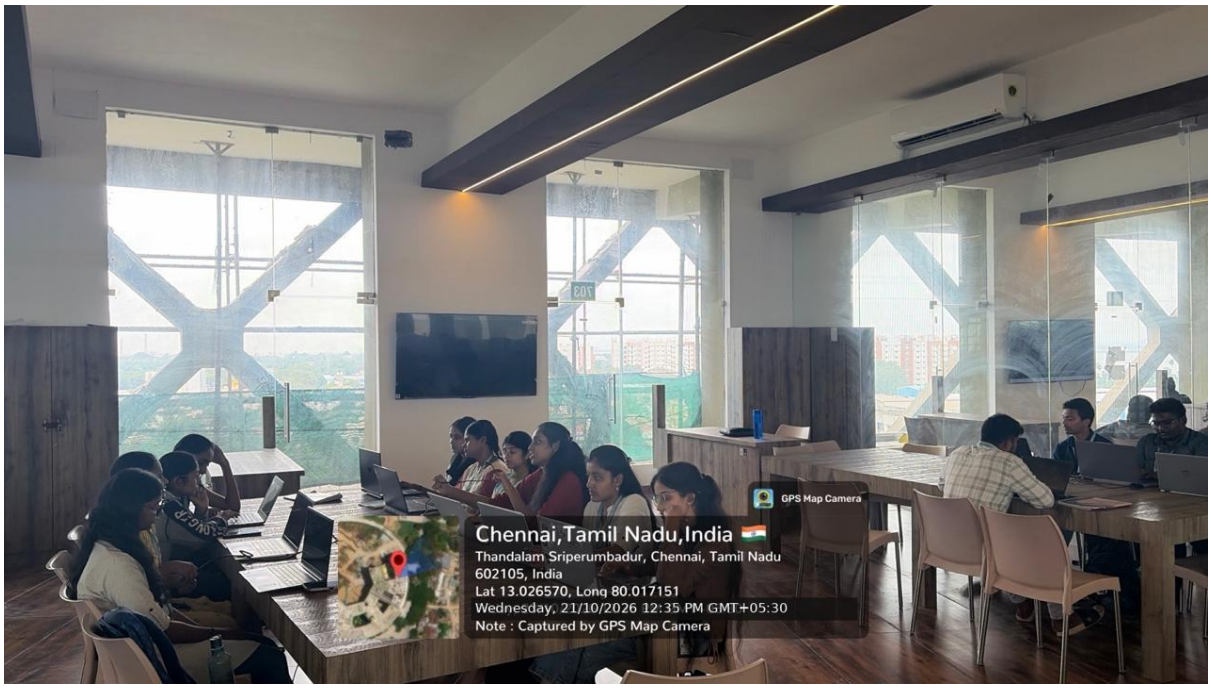
3. Region-Based Segmentation Integration: Edge boundaries provide structural constraints for region-growing algorithms and active contour models (Snakes). They effectively isolate distinct regions, allowing object measurement, feature extraction, and automated scene analysis.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____





The following assessment has been discussed by the faculty and done at the class on 21st October 2026.